Multi-Objective Green Multimodal Transport Path Optimization with the Participation of High-Speed Railway

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***Abstract*—The development of green environmental protection concept puts forward higher requirements for transportation. The superior performance of high-speed railway makes it an inevitable trend to participate in the combined transportation of goods. In this paper, a multi-objective green multimodal transportation path optimization model is established by considering the restriction of node schedule and customer receiving time window. This model combines with three transportation modes of roads, waterway and high-speed railway. It aims to minimize carbon emissions and transportation costs while maximizing logistics service satisfaction. Non-dominated sorting genetic algorithm based on reference points (NSGA-III) is applied to solve the model. And the feasibility and effectiveness of the model and algorithm are verified by a numerical example, which provides a variety of transportation schemes to meet the differentiated needs of customers for multimodal transport operators.**

Keywords—high-speed rail transportation； multi - objective model； carbon emissions； multimodal transport； NSGA-III

I. INTRODUCTION

With the promotion of green logistics development concept, enterprises have put forward higher requirements for timeliness, economical efficiency and service quality of multimodal transport. As a modern transportation mode, high-speed railway has the characteristics of less time-consuming, low energy consumption, and high safety. Therefore, it is of great significance to take high-speed railway to freight transport and promote the development of "high-speed rail+" transport mode for multimodal transport practice.

At present, some scholars have studied the multimodal transport model. For example, Dulebenets et al. [1] constructed a minimization model of combined transportation cost and decay cost by considering product characteristics. Zhao Chunlei et al. [2] established a three-stage multimodal transportation model with cloud delivery demand, it aims to minimize total cost. Li Mengliang et al. [3] constructed a total cost minimization model, aiming to minimize transportation hub location cost and transportation cost in multimodal transportation network. Liu Song et al. [4] constructed a minimization model of transportation and transit costs by considering arrival time interval. In recent years, carbon emission control has also become an important goal of multimodal transport. Such as, Li Huifang et al. [5] constructed an optimization model of multimodal transport path by considering multi-task carbon emissions. Liu Jie et al. [6] built a multi-objective optimization model to minimize the total transportation cost and carbon emission, which is solved by NSGA-II. Li Jun et al. [7] provided a green multimodal transportation path optimization model that took into account transportation time, transit time, customer demand and transportation volume. However, most of the multimodal network modeling only consider roads, ordinary freight railways and waterways nowadays. And the researches on multimodal transport network with high-speed railway mainly focus on railway planning and design. There are few researches on multimodal transport path planning. For example, the adaptability analysis of high-speed railway network by considering the capacity of different high-speed railway lines [8]. The connection design of high-speed railway station and bus [9]. And the minimum total distribution cost of fresh agricultural products with the participation of high-speed railway [10]. In fact, the time factor has an important influence on the route selection for the multimodal transport network involved by high-speed railway, as it does not carry bulk cargo transportation like ordinary railway and has very strict on the arriving and departing time.

Therefore, based on the multimodal transport network composed of roads, waterway and high-speed railways, a multi-objective green multimodal transport path optimization model is built with considers the constraints of hard time window of ships and high-speed railways and customer soft time window at each transport node in this paper. It aims to minimize multiple costs while maximizing logistics service satisfaction. Then a multi-objective optimization algorithm named NSGA-III is applied to solve the model for providing multiple decision plans to multimodal transport operators.

II. PROBLEM DESCRIPTION AND MODELING

A. Problem Description and Model Assumptions

Firstly, a multimodal transport network is constructed by considering the schedule restrictions of different transportation modes at each node and the time window

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constraints of customers receiving goods. It is assumed that a number of container goods need to be transported from city *O* to city *D* in this network. Between the two cities, there are a number of transit nodes. Three transportation modes can be chosen between any adjacent nodes including roads, waterway and high-speed rail. In addition to road transportation, other transportation modes at each node have different periodic idle time windows. If the goods arrive beyond the free time window, it only can be transferred in the next free time window at the node. In the process of multimodal transport, the following assumptions are made:

(i) The same batch of goods cannot be split during transportation. There is no conversion between the same mode of transportation;

(ii) The transfer and replacement operation only occurs at the node. Only one mode of transportation can be selected between any adjacent nodes. The same node can transfer at most once;

(iii) In the process of intermodal transport, only calculated the carbon emissions in the process of freight transport and transfer loading.

B. Parameters Description

O represents departure city; D represents destination city; **W** is a collection of node city; **M** is a collection of transportation modes; Q represents the amount of carrier container.

Transport cost variables: C_r^k is the base price for transport in mode k; C_s^k is the freight rate for transport in mode k; c_j^{kl} is the unit transfer cost when the mode of transport changes from k to l at node j; c_j^l is the unit time cost of waiting mode l for transshipment at node j; d_{ij}^k represents the transportation mileage from node i to node j in mode k.

Time variables: T_{wj}^l is the time to wait for transport in mode l at node j; T_{gj}^l is the time to wait to leave in mode l at node j; T_{ij}^k represents the transportation time taken from node i to node j in mode k; T_j^{kl} is the transit time when the mode of transportation changes from k to l at node j; T_i represents the actual time when the goods arrive at node i; T_D represents the transportation time for the goods to reach the destination city D; C_E represents the storage cost per unit time for the goods arriving at the destination in advance; C_L represents the penalty costs per unit time for the goods arriving late at the destination; $[T_E, T_L]$ represents the customer receiving time window; $[V_E^k, V_L^k]$ represents the Idle time window of transport mode k at the current node; $[V_E^{k*}, V_L^{k*}]$ represents the next idle time window of transport mode k at the current node.

Carbon emission variables: ρ represents carbon tax value (yuan/kg); B_{ij}^k represents the unit carbon emissions transported from node i to node j in mode k; B_j^{kl} represents the carbon emissions when the transportation mode is changed from k to l at node j.

Satisfaction variables: o_{ij} is the logistics service quality between node i and node j solved by the fuzzy grade evaluation method; H_T is the timeliness evaluation index of multimodal transport; H_C is the economic evaluation index of multimodal transport; H_N is the service evaluation index of multimodal transport; C_{min} represents the minimum total transportation cost; C_{max} represents the maximum total transportation cost.

Decision variables: x_{ij}^k represents whether mode k is used for transportation from node i to node j, the value is 1, otherwise it is 0; y_j^{kl} represents if the mode of transport changes from k to l at node j, the value is 1, otherwise it is 0.

C. Model Construction

$$\min Z_1 = Q\left\{\sum_{i,j\in W}\sum_{k\in M}(C_r^k + C_s^k d_{ij}^k)x_{ij}^k + \sum_{j\in W}\sum_{k,l\in M} c_j^{kl} y_j^{kl} + \sum_{j\in W}\sum_{k,l\in M} c_j^l T_{wj}^l y_j^{kl}\right\} + Q\left\{C_E \max(T_E - T_D, 0) + C_L \max(T_D - T_L, 0)\right\} \quad (1)$$

$$\min Z_2 = Q\left(\sum_{i,j\in W}\sum_{k\in M} B_{ij}^k d_{ij}^k x_{ij}^k + \sum_{j\in W}\sum_{k,l\in M} B_j^{kl} y_j^{kl}\right) \quad (2)$$

$$\max Z_3 = H_T + H_C + H_N \quad (3)$$

S.T.

$$\max H_T = \begin{cases} \frac{1}{2} + \frac{1}{2}\sin\frac{\pi}{T_E}[T_D - \frac{T_E}{2}] & ,0 < T_D < T_E \\ 1 & ,T_D = T_E \\ \frac{1}{2} - \frac{1}{2}\sin\frac{\pi}{T_L - T_E}[T_D - \frac{T_E + T_L}{2}] & ,T_E < T_D \le T_L \\ 0 & ,T_L < T_D \end{cases} \quad (4)$$

$$\max H_c = \begin{cases} 1 & ,Z < C_{\min} \\ 1 - 2(\frac{Z - C_{\min}}{C_{\max} - C_{\min}})^2 & ,C_{\min} < Z \le \frac{C_{\min} + C_{\max}}{2} \\ 2(\frac{Z - C_{\max}}{C_{\max} - C_{\min}})^2 & ,\frac{C_{\min} + C_{\max}}{2} < Z \le C_{\max} \\ 0 & ,C_{\max} < Z \end{cases} \quad (5)$$

$$\max H_N = \frac{\sum_{i,j\in W}\sum_{k\in M} x_{ij}^k o_{ij}}{\sum_{i,j\in W}\sum_{k\in M} x_{ij}^k} \quad (6)$$

$$T_D = \sum_{i,j\in W}\sum_{k\in M} T_{ij}^k x_{ij}^k + \sum_{j\in W}\sum_{k,l\in M} T_j^{kl} y_j^{kl} + \sum_{j\in W}\sum_{k,l\in M} T_{wj}^k y_j^{kl} + \sum_{j\in W}\sum_{k,l\in M} T_{gj}^l y_j^{kl} \quad (7)$$

$$Z = Z_1 + \rho Z_2 \quad (8)$$

$$\sum_{k\in M} x_{ij}^k \le 1 \quad \forall i,j \in W \quad (9)$$

$$\sum_{k,l\in M} y_j^{kl} \le 1 \quad \forall j \in W \quad (10)$$

$$x_{ij}^k + x_{ji}^l \ge 2y_j^{kl} \quad \forall i,j \in W, \forall k,l \in M \quad (11)$$

$$\sum_{k\in M}\sum_{i\in W} x_{ij}^k - \sum_{k\in M}\sum_{i\in W} x_{ji}^k = \begin{cases} -1 & ,j = O \\ 1 & ,j = D \\ 0 & ,j \ne O,D \end{cases} \quad (12)$$

$$x_{ij}^k, y_j^{kl} \in \{0,1\} \quad \forall i,j \in W, \forall k,l \in M \quad (13)$$

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$$T_{wj}^{k}=\begin{cases}V_{E}^{k}-T_{i} & ,T_{i}<V_{E}^{k}\\ 0 & ,V_{E}^{k}\le T_{i}<T_{i}+T_{i}^{kl}\le V_{L}^{k}\ \ \forall k,l\in M,\forall i,j\in W\\ V_{E}^{k*}-T_{i} & ,V_{L}^{k}<T_{i}+T_{i}^{kl}\end{cases}\quad(14)$$

$$T_{gj}^{l}=\begin{cases}V_{L}^{k}-V_{E}^{k}-T_{i}^{kl} & ,T_{i}<V_{E}^{k}\\ V_{L}^{k}-T_{i}-T_{i}^{kl} & ,V_{E}^{k}\le T_{i}<T_{i}+T_{i}^{kl}\le l_{ij}^{k}\ \ \forall k,l\in M,\forall i,j\in W\\ V_{L}^{k*}-V_{E}^{k*}-T_{i}^{kl} & ,V_{L}^{k}<T_{i}+T_{i}^{kl}\end{cases}\quad(15)$$

Equation (1) represents the transportation cost, which is composed of the transportation cost in the transportation process, the transfer loading cost, the storage cost and the penalty cost while the goods are waiting for transshipment. Equation (2) represents carbon emissions, which is composed of carbon emissions in the transport process and carbon emissions in the transit process. Equation (3) represents the logistics service satisfaction, which is composed of the timeliness index of multimodal transport, the economic index of multimodal transport and the service index of multimodal transport. Equation (4) represents the timeliness evaluation index of multimodal transport, and the relationship between it and transport time is described by the combination distribution of the rising ridge shape and the falling ridge shape functions in the fuzzy theory. Equation (5) represents the economic evaluation index of multimodal transport, and the relationship between it and the total transportation cost is described by Z function in the fuzzy theory. Equation (6) represents the service evaluation index of multimodal transport, which is related to the infrastructure level, economic development level and logistics service level of each node. It is evaluated by the fuzzy grade evaluation method. Equation (7) represents the total transport time, consisting of time spent in the transport process, time spent in the transshipment process, time spent waiting for transshipment at the node and time spent waiting for departure at the node. Equation (8) represents the total logistics cost, which is composed of transportation cost and carbon emission cost under the condition of considering carbon tax value. Equation (9) represents that only one transport mode can be chosen between node *i* and node *j*. Equation (10) represents that at most one transshipment occurs at node *i*. Equation (11) ensures that the whole transportation process is continuous. Equation (12) ensures the flow conservation of node. Equation (13) represents that the decision variables are 0-1 variables. Equation (14) indicates the waiting time for transfer at node *j*. Equation (15) indicates that it takes time to wait for departure at node *j*.

III. ALGORITHM DESIGN

The model is a multi-objective optimization model with three targets. NSGA-III algorithm has a better ability for solving multi-objective optimization problems with more than two objectives [11]. The main principle of NSGA-III algorithm is to perform fast non-dominated sorting on the objective function value of each individual, and divide the number of hierarchical levels for the individua. Parent population is selected according to the non-dominant relationship and the distance of the associated reference points. Then offspring population is produced by crossing and mutation. A new population is produced by merging the parent population with the offspring population. The above non-dominated operation would be carried out repeatedly. The mainly steps is as follows:

(i) Encoding and decoding strategy based on real number priority

Most of the research uses real integer coding combined with other coding methods for the multimodal transport. One coding section represents the route of transportation, and the other coding sections represents the mode of transportation. This coding method makes chromosomes long and complicated to decode, cross and mutate. In this paper, a real number coding method based on priority is designed to encode transportation schemes. In this coding method, the gene sequence not only represents the priority of the transportation route but also represents the transportation mode at each node. It can reduce the operation complexity.

The operations can be describe as follows: Firstly, generate several unrepeated random numbers in the numerical range of [1, 4). Each value of the random number is related to a transportation mode at the node. Here we set the integer part of each number to express a transportation mode, e.g. 1 means road transportation, 2 is water transportation, and 3 is high-speed rail transportation. Take Fig. 1 as an example. In the multimodal transport network, there are three available transportation modes and six nodes including the starting point and the end point. The value range of each gene in chromosome is [1, 4).

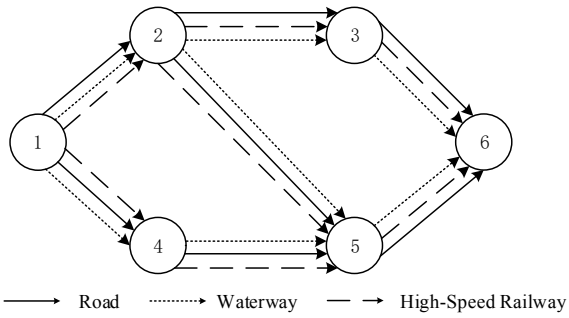


Fig. 1. Multimodal transport network diagram

As the coding expression shown in Fig. 2, firstly, the starting node is fixed as node 1. According to the relationship table between nodes which is given in TABLE I, the reachable nodes of node 1 include node 2 and 4. In the chromosome, the priority of node 2 is 3.46, and the priority of node 4 is 1.54. Therefore, the next arrival node from node 1 to is node 2, and the path 1→2 is selected. At the same time, it is also confirmed that the transportation mode 3 is used between the two nodes. And so on, until the node 6 is reached, the decoding operation is completed. Finally, a multimodal transport plan got from the above chromosome is 1→2→3→6. The mode of transportation between nodes is 3—3—3.

Transport node number

1	2	3	4	5	6
1.28	3.46	3.17	1.54	2.12	3.25

Chromosomes

Fig. 2. Expression of a chromosome

TABLE I. RELATIONSHIP TABLE BETWEEN NODES

Current Node	Reachable Node
1	2, 4

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2	3, 5
3	6
4	5
5	6
6	—

(ii) Non-dominated sorting strategy

This strategy is provided to improve the classic operation. Firstly, it calculates the objective function value of each individual in the population and normalizes them. A non-dominated sorting method is applied to divide the population into several different non-dominated layers. Then according to the rank order of the non-dominated stratum and the relevance of the reference point, the solutions of each non-dominated layer are added to a new population layer by layer, until all the individuals are handled.

The ranking order of each individual is based on the number of individuals that the individual dominates and the number of individuals that dominate the individual. Associating reference point distances requires to calculate the distance from each individual to the ideal reference point. It need to determine the correlation between the individual and the reference point. Then according to the number of individuals associated with the reference point, the individual which is the closest one to the reference point is selected to join in the population.

(iii) Chromosome crossover and variation strategy

In order to improve the searchability of the algorithm, a simulated binary crossover operator is used. The crossover distribution index is set as 20. Similarly, in order to improve the diversity, the polynomial mutation operator is applied to carry out mutation operation. The variation distribution index is set as 20.

IV. Example Design and Result Analysis

A. Case Design

A case from practice is designed to test effectiveness of the model and algorithm. It is assumed that there are twenty 20ft container need to be transported from city O to city D. During the process, it needs to pass through seven urban nodes. and three transportation modes can be chosen between any adjacent nodes including roads, waterway and high-speed rail. In addition to road transport, other modes of transport have different schedule restrictions. Customer delivery time window is [25, 40]. The storage cost per unit time is 600 yuan/(TEU·h) when the goods arrives earlier than the customer earliest delivery time. The penalty cost per unit time is 1200 yuan/(TEU·h) when the goods arrives late at the customer latest receipt time. The carbon tax rate is 20 yuan/t. The multimodal transport network is shown in Fig. 3. Among them, the evaluation indicators of multimodal transport service include node infrastructure construction, economic development level and logistics service level. The corresponding weight coefficients of each factor are {0.2, 0.2, 0.6}. The rating is divided into {good, general, bad}. The corresponding score is set to {0.9, 0.6, 0.3} [12]. Then nine groups of node service evaluation data are randomly generated and calculated based on a fuzzy hierarchy method.

In this multimodal network, the transport base and transport prices for various modes of transport are shown in TABLE II. The waiting costs and unit carbon emissions for transport modes at nodes are shown in TABLE III. The transportation time and distance of each transportation path corresponding to different transportation modes are shown in TABLE IV. The unit transshipment costs, transshipment times, and unit transshipment carbon emissions between transport modes at the node are shown in TABLE V. The time window limits for waterways and high-speed rail at each node are shown in TABLE VI.

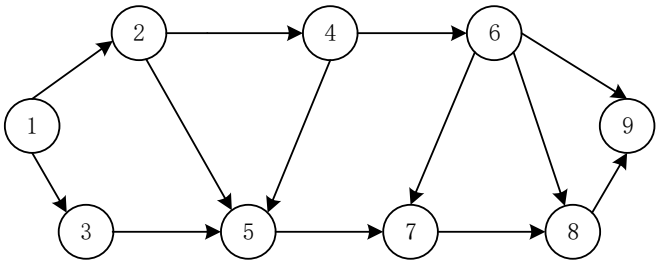


Fig. 3. Multimodal transportation network diagram

TABLE II. The Basic Transportation Price and Freight Rate of Each Transportation Method

Transport Mode	Road	Waterway	High-Speed Railway
Base price (yuan/TEU)	20	8	260
Freight rate {yuan/(TEU·km)}	3	0.429	1.29

TABLE III. Waiting Costs Per Unit Time and Unit Carbon Emissions of Each Transportation Mode at The Node

Transport Mode	Road	Waterway	High-Speed Railway
Waiting cost per unit time {yuan/(TEU·h)}	—	40	100
Unit carbon emissions {kg/(TEU·km)}	0.9	0.328	0.198

TABLE IV. Transportation Time and Transportation Distance Under Different Transportation Modes of Each Transportation Route

Transport Path	Road		Waterway		High-Speed Railway	
	Time (h)	Distance (km)	Time (h)	Distance (km)	Time (h)	Distance (km)
(1,2)	5	350	12	480	2	460
(1,3)	4	320	11	450	2	480
(2,4)	5	400	14	560	2	460
(2,5)	4	300	9	360	1	250
(3,5)	6	480	10	400	2	500
(4,5)	4	300	9	360	1	250
(4,6)	6	450	13	520	2	460
(5,7)	6	470	18	720	2	500
(6,7)	4	330	11	450	2	400
(6,8)	6	460	11	455	2	480
(6,9)	6	500	16	640	2	520

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TABLE V. UNIT TRANSFER COST, TRANSFER TIME AND UNIT TRANSFER CARBON EMISSIONS AMONG VARIOUS TRANSPORTATION MODES AT NODES

Transport Path	Road			Waterway			High-Speed Railway		
	Transshipment cost (yuan/TEU)	Transit time (h)	Carbon emission (kg/TEU)	Transshipment cost (yuan/TEU)	Transit time (h)	Carbon emission (kg/TEU)	Transshipment cost (yuan/TEU)	Transit time (h)	Carbon emission (kg/TEU)
Road	—	—	—	160	0.5	16	240	0.8	12
Waterway	160	0.5	16	—	—	—	280	1.2	20
High-speed railway	240	0.8	12	280	1.2	20	—	—	—

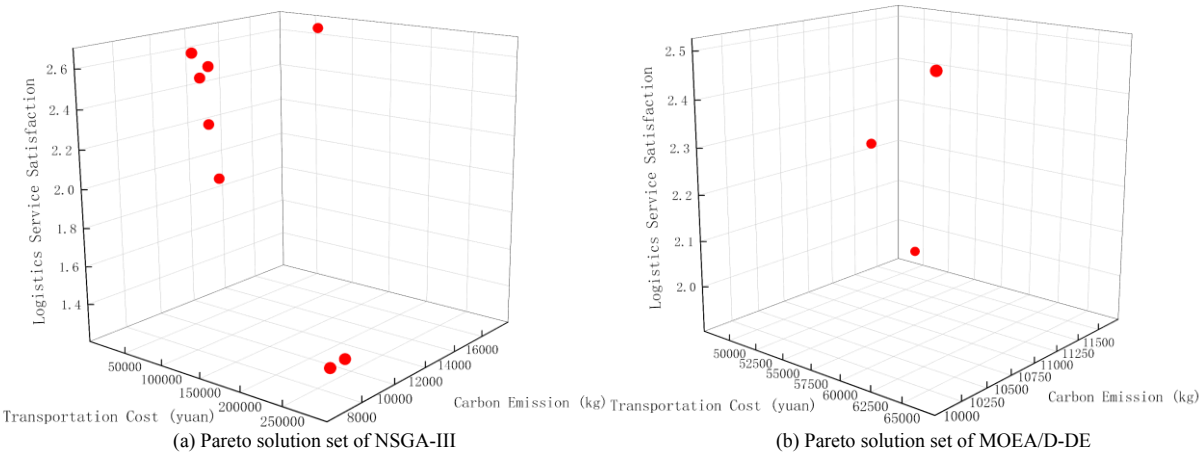
TABLE VI. SCHEDULE RESTRICTIONS OF VARIOUS TRANSPORTATION MODES AT VARIOUS NODES

Transport Mode	Time window					Transit node
Waterway transportation time window	[5,8]	[10,13]	[15,18]	[20,23]		2、3、4、7、8
	[2,6]	[8,14]	[16,20]	[21,23]		5、6
High-speed rail transportation time window	[1,3]	[5,7]	[9,11]	[14,16]	[18,20]	2、4、7、8
	[0,2]	[5,7]	[10,12]	[14,16]	[18,20]	3
	[2,5]	[7,10]	[12,14]	[16,18]	[21,23]	5、6

（7,8）	4	320	11	450	2	400
（8,9）	6	500	16	600	2	520

B. *Result analysis*

The population size is set as 100. The maximum number of iterations as 100. The initial mutation probability as 0.1. And the crossover probability is 0.9. NSGA-III algorithm and



(a) Pareto solution set of NSGA-III (b) Pareto solution set of MOEA/D-DE

Fig. 4. Spatial distribution diagram of Pareto solution set

TABLE VII. THE OPTIMAL PARETO SOLUTION SET FOR EACH OBJECTIVE

Transportation Plan	Transportation Cost(yuan)	Carbon Emission(kg)	Logistics Service Satisfaction	Transportation Time(h)	Transportation Route	Transportation Mode
1	50009	11653	1.943	35	1-2-4-6-9	3-3-2-2
2	50527	11103	2.245	32	1-2-4-6-9	2-2-3-3
3	64855	10063	2.514	29	1-2-4-6-9	3-3-3-2
4	73717	10176	2.574	28	1-2-4-6-9	3-2-3-3
5	74630	9251	2.658	26	1-2-4-6-9	2-3-3-3
6	77958	16774	2.662	25.5	1-2-4-6-9	3-3-2-1
7	272954	8435	1.358	9	1-2-5-7-8-9	3-3-3-3-3
8	273820	7524	1.354	8	1-2-4-6-9	3-3-3-3

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MOEA/D-DE algorithm are selected to solve the constructed model respectively. The distribution diagram of the non-dominated solution sets got by the two compared algorithms are shown in Fig. 4. The pareto solution set obtained by NSGA-III algorithm is shown in TABLE VII. Comparing the spatial distribution of pareto solution set obtained by the two algorithms, it can be seen that the number of non-dominated solutions obtained by the NSGA-III algorithm is larger and the distribution interval is wider. NSGA-III algorithm shows a good multi-target search performance. The algorithm is effective for solving the model.

It can be seen from TABLE VII that the containers are transported by combined transportation in most of the pareto frontier solutions. And the frequency of high-speed rail introduced into the optimal plans is much higher than that of roads. Mainly, it because of that although the transportation base price of road transportation is low, but the transportation price is high compared with high-speed rail transportation. The transportation cost of using road transportation is higher than high-speed rail transportation When the transportation distance is longer. And road transport has higher carbon emissions and longer transport time. Therefore, when the customer's time window is small, it is more economical and efficient to choose high-speed rail than road.

In summary, with the increasing times that high-speed railway participates in freight transportation, transportation cost shows an upward trend. Carbon emissions and transportation time are declining. And the logistics service satisfaction rises firstly and then declines. It mainly because of the participation of high-speed rail transportation which makes the total transportation time is constantly closer to the earliest delivery time window of customers and the carbon emissions is reduces. Therefore, within a certain range of changes in transportation costs, the optimal transportation routing remains unchanged, and the logistics service satisfaction keeps rising.

However, when goods are transported by high-speed rail alone, not only the transportation cost is increasing, but also the reduction in transportation time and carbon emissions does not bring better economic benefits. Instead, it leads to a large storage costs, decreased logistics services satisfaction, and changes in transportation routes. Therefore compared with a single mode of transportation, it is better to promote logistics to reduce costs and increase benefits by reasonably using the "high-speed rail +" combined transport mode.

V. CONCLUSION

In order to satisfy the requirement of the current development trend of green transportation, a multi-objective green multimodal transport model is proposed which considers transportation costs, carbon emissions, and logistics service satisfaction. From the perspective of multimodal transport operators, the schedule restrictions of different transportation modes at each node and the time window restrictions of customers receiving goods are considered. The NSGA-III algorithm is introduced to solve the model. The feasibility of the model and algorithm is verified and the problem can be solved effectively. The results show that the combined transportation mode, which combines high-speed railway with other transportation, has obvious advantages to reduce logistics transportation cost, shorten transportation time and improve logistics service satisfaction, so as to provide multimodal transportation operators with more transportation solutions to meet the different needs of customers.

In future, the influence of transport capacity and customer time window changes on multimodal transport path optimization will be studied, and resource splitting and allocation at nodes caused by customer demand changes also would be analyzed.

REFERENCES

[1] M. A. Dulebenets, E. E. Ozguven, R. Moses, and M. B. Ulak, "Intermodal freight network design for transport of perishable products," Open Journal of Optimization, vol. 5, pp. 120–139, June 2016.

[2] Zhao Chunlei, Meng Yabin, Liu Jianbo, Xiong Wanhong, and Qiao Peng, "Dispatch model and solution strategy for multimodal transportation based on logistics cloud platform," Journal of the China Railway Society, vol. 40, pp. 1–8, January 2018.

[3] Li Mengliang, Wang Xifu, Sun Quanxin, and Wang De. "Hub location model and algorithm for road-rail intermodal transportation network," Journal of the China Railway Society, vol. 38, pp. 1–7, December 2016.

[4] Liu Song, Peng Yong, Song Qiankun, and Zhong Yiying, "The robust shortest path problem for multimodal transportation considering timetable with interval data," Systems Science & Control Engineering, vol. 6, pp. 68–78, July 2018.

[5] Li Huifang and Su Luan, "Multimodal transport path optimization model and algorithm considering carbon emission multitask," Journal of Supercomputing, vol. 76, pp. 9355–9373, February 2020.

[6] Liu Jie, Peng Qiyuan, and Yin Yong, "Multimodal transportation route planning under low carbon emissions background," Journal of Transportation Systems Engineering and Information Technology, vol. 18, pp. 243–249, December 2018.

[7] Li Jun, Yang Bin, and Zhu Xiaolin, "Path optimization of green multimodal transportation under mixed uncertainties," Journal of Transportation Systems Engineering and Information Technology, vol. 19, pp. 13–19, August 2019.

[8] Bi Mingkai, He Shiwei, and Xu Wangtu (Ato), "Express delivery with high-speed railway: Definitely feasible or just a publicity stunt," Transportation Research Part A, vol. 120, pp. 165–187, February 2019.

[9] An Jiuyu, Song Rui, Bi Mingkai, and Xue Shouqiang, "Optimization flexible route design for high-speed railway station feeder bus," Journal of Transportation Systems Engineering and Information Technology, vol. 19, pp. 150–155, October 2019.

[10] Li Yumin, Qiu Meng, Yan Kaili, and Liu Qixin, "Multimodal transportation route selection of fresh products considering time window with the participation of high-speed rail," Journal of Chongqing Jiaotong University (Natural Sciences), vol. 40, pp. 54–61, April 2021.

[11] Liu Jianchang, Li Fei, Wang Honghai, and Li Tianjun, "Survey on evolutionary many-objective optimization algorithms," Control and Decision, vol. 33, pp. 879–887, January 2018.

[12] Wan Jie and Wei Shuang, "Multi-objective multimodal transportation path selection based on hybrid algorithm," Journal of Tianjin University, vol. 52, pp. 285–292, January 2019.

[13] Zhang Zhenjiang, Zhang Yuzhao, and Wang Xiaorong, "Low-carbon transportation mode and path selection of express freight under fuzzy demand condition," Journal of Railway Science and Engineering, vol. 18, pp. 777–785, March 2021.

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